Kavyanjali Agnihotri

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Education

University of Tübingen Tübingen, Germany
M.Sc. in Machine Learning, Grade: 2.18 (German scale: 1.0 best, 4.0 pass) *Oct 2024 – Mar 2027 (expected)*

Birla Institute of Technology & Science (BITS) Pilani Hyderabad, India
B.E. in Computer Science, CGPA: 8.49/10 (10 best; German-scale equivalent: 1.8) *Nov 2020 – Jul 2024*

Research Experience

Automated Defence Discovery against Prompt Injection Master's Thesis, University of Tübingen
Supervisors: Prof. Andreas Geiger, Dr. Jonas Geiping *Jul 2026 – Present*

- *Question:* Defences tested on fixed attacks can look safe without being safe (Meta-SecAlign-8B reports ∼0% attack success, yet gradient attacks break it). Can LLM agents find defences that survive new attacks?
- *Pilot:* An LLM agent tuning Llama Guard 3 raised its benchmark score by gaming the metric: fewer false alarms, but fewer harmful prompts caught (75% → 71%).
- *Attacks:* Reproduced 14 gradient attacks on Meta-SecAlign-8B; their success fell from 21/25 to 2–3/25 when the goal was a realistic action, such as a phishing link, instead of a fixed string.
- *Method:* Re-implemented four defences as baselines; building a loop where the agent writes its own defences.

Skill-File Injection in Open-Weight LLM Agents Team project
Python, vLLM, OpenHands *Jun 2026*

- Evaluated 9 open-weight models (up to 122B) in 540 sandboxed agent runs; attack success ranged from 3% to 46%.
- Showed LLM judges miss executed malicious scripts; a command-history check found up to 43% more attacks.

Compositional Understanding in Stable Diffusion Research Project
Python, PyTorch, Stable Diffusion 3.5 *Sep 2025 – Mar 2026*

- Swapped SD3.5's CLIP text encoder for a compositionally fine-tuned one: "above" accuracy rose from 73% to 90%.
- Built a 1,300-prompt spatial-relation benchmark with a vision-language model judge (73% human agreement).

Workload Analysis of ML GPU Clusters Team Project
Python, pandas, scikit-learn *Nov 2025 – Feb 2026*

- Analysed 2.4M jobs from two university GPU clusters: users used only 3–7% of the run time they requested, and 75% of jobs on the H100 cluster requested no GPU.

Reinforcement Learning for Air Hockey Course Project
PyTorch, TD3 *Jan 2026 – Feb 2026*

- Trained a TD3 agent with self-play, prioritised replay and pink noise: 80% win rate vs. the strong built-in opponent.

Work Experience

Student Research Assistant, Scholar Inbox University of Tübingen
Python, llama.cpp, Celery, Flask, React *Dec 2024 – Present*

- Benchmarked llama.cpp vs. vLLM; moved the daily paper-QA pipeline to a quantised Qwen3.6-35B on one GPU.
- Built a pipeline that turns papers into podcasts, with a 9.3× speedup from parallel task queues and projected audio storage cut from 4.8 GB to 160 MB per day.
- Built a dashboard for versioning and testing production LLM prompts, a rate limiter for a paid text-to-speech API, and user features such as a podcast player.

Technical Intern, Cisco Bengaluru, India
Python, Docker, Jenkins, Java *Jun 2023 – Jul 2023, Jan 2024 – Jun 2024*

- Containerised 8+ repositories, set up CI/CD, and upgraded vulnerable Java libraries in 20+ microservices.
- Built a real-time view of runtime errors in the Secure Network Analytics interface.

Achievements, Teaching & Languages

Zuse School ELIZA scholarship: 24-month stipend for M.Sc. studies Oct 2024 – Sep 2026
Summer schools: MLSS (ML Reliability & Safety), Kraków; IICCSSS (Cognitive Science), Darmstadt 2025 – 2026
Hackathons: Google Girl Hackathon, top 2.5%; JPMorgan Code for Good, finalist of 25+ teams 2023
Teaching & volunteering: TA for Human-Computer Interaction, BITS Pilani; volunteer at MLSS Tübingen 2026
Programming: Python, C++, C, Java, JavaScript/TypeScript, SQL, Bash
Languages: English (fluent), Hindi (native), German (A2)